THE UK UNIVERSITY INTEGRATION BEE 2025/2026

$$\int$$

ROUND TWO

CROSSNUMBER ROUND

DOWN

Sponsored by



DOWN

1. A divisor of $5\displaystyle\int_{15}^{80}\sqrt{\sqrt{2x-2\sqrt{x^2-1}}+\sqrt{x-1}}\,\mathrm{d}x$.

2. The second-largest possible three-digit integer value of ka, where $k\in\mathbb{Z}$ and a is an integer such that the following statement is true:

   There exists one and only one $x\in\mathbb{R}$ such that both $ax-2026<x^2-ax$ and, for every integrable function $f:\mathbb{R}\to\mathbb{R}$, the average value of f on $[ax-2026,x^2-ax]$ is $\displaystyle\int_{ax-2026}^{x^2-ax}f(t)\,\mathrm{d}t$.

3. Three consecutive terms (all non-negative integers) of a geometric sequence put together and read as a single number.

4. A four-digit number $\overline{abcd}$ such that $\displaystyle\int_0^1\frac{\tanh^{-1}\left(\sqrt{1-x^{\frac{a}{b}}}\right)}{1-x}\,\mathrm{d}x=\frac{c\pi^k}{d}$, where k is some rational number.

6. A prime factor of (10 DOWN). Also, if you reverse the digits of (6 DOWN), it's still prime!

8. An unhappy prime.

10. Johannes Kepler was a prominent astronomer and mathematician. In his work *Nova stereometria doliorum vinariorum*, he calculated volumes of solids of revolution using infinitesimal techniques. This work, based on wine barrels of all things, formed the basis of integral calculus and is considered one of the most significant works in the prehistory of calculus.

    The answer to this clue is greater than Kepler's age when he died.

13. $\left\lfloor\dfrac{\text{(13 DOWN)}}{10}\right\rfloor=\displaystyle\int_{-6}^{6}f(x)\,\mathrm{d}x$, where f is an integrable 4-periodic function on $[-6,6]$ satisfying the equations

$$2\int_2^6 f(x)\,\mathrm{d}x+6\int_5^6 f(x)\,\mathrm{d}x=26\qquad\text{and}\qquad 5\int_2^5 f(x)\,\mathrm{d}x+2\int_5^6 f(x)\,\mathrm{d}x=25.$$

14. The largest positive integer n such that

$$\left\lceil\frac{1}{\pi}\int_0^{2^n\pi}\underbrace{\sqrt{2+\sqrt{2+\sqrt{\cdots\sqrt{2+2\cos(x)}}}}}_{n\text{ radicals}}\,\mathrm{d}x\right\rceil$$

    has exactly (3 ACROSS) + 1 digits when expressed in hexadecimal (base 16).

16. A two-digit number $\overline{ab}$ such that $\displaystyle\int_1^\infty\frac{(x+2)\ln(x)}{x^k+x^3+x}\,\mathrm{d}x=\frac{\sqrt[3]{\pi^a}}{b+k}$, where k is the smallest prime factor of (3 ACROSS).

20. A three-digit number $\overline{abc}$ such that $\left|\displaystyle\int_0^1\frac{\ln\left(x^8-x^7+x^5-x^4+x^3-x+1\right)}{x}\,\mathrm{d}x\right|=\frac{a\pi^2}{bc}$.

In the answers to the following clues, nine of the digits 0-9 appear once and exactly once, and one digit does not appear at all.

17. The product of any two digits of this number is greater than the remaining digit.

18. This number is 5 more than a square number.

19. The first digit, multiplied by the second digit, gives a two-digit number $\overline{ab}$ with $a - b$ being the last digit of (18 DOWN).